Dynamic Panels

BUSS975: Causal Inference in Financial Research

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Where we left off

- The panel lecture rested on one claim: anything about a firm that **does not change** cannot explain why its outcome changed. Demean, and the firm effect is gone.
- The price was **strict exogeneity**: today's shock must not feed into tomorrow's regressors.

- Finance is full of outcomes that depend on **their own past**. Leverage, payout, cash, investment — each is where it is largely because of where it was last year.
- Put yesterday's Y on the right-hand side and strict exogeneity is not merely doubtful. **It is false by construction.**
- This lecture: what breaks, how badly, how to fix it — and when the fix does not work.

The plan

1. **Why the past belongs on the right-hand side** — and one rule that settles every calculation we will need.
2. **Two estimators, two wrong answers.** They err in opposite directions, which turns out to be useful.
3. **The fix: instruments from the past.** Anderson–Hsiao, then Arellano–Bond.
4. **Why the fix can fail**, and what people do then.
5. **In practice:** a case study from capital structure, the code, and a checklist.

- Two running questions. *Which comparison is this estimator making?* And, new this week: *which way is it wrong?*

1. Why the Past Belongs on the Right-Hand Side

Most things in finance drift

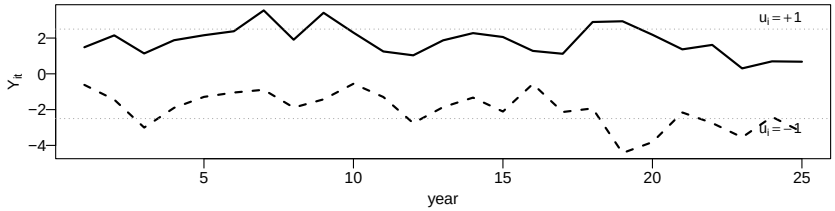
They do not jump to a target. They close part of the gap each year.

Leverage	firms close only part of the gap to target each year
Payout	managers smooth dividends, and cut only as a last resort
Cash	drawn down quickly, rebuilt slowly

$$Y_{it} \;=\; \rho\, Y_{i,t-1} + \delta D_{it} + u_i + \varepsilon_{it}, \qquad |\rho| < 1$$

- ρ is **not a nuisance parameter — it is the economics.** $1-\rho$ is the share of the gap closed per year; a shock has half-life $\ln(0.5)/\ln\rho$.
- $\rho = 0.6$: close 40% a year, half-life 1.4 years. $\rho = 0.9$: half-life 6.6 years. **Those are different theories of the firm**, and $\hat{\rho}$ is supposed to tell them apart.
- δ is still the treatment effect, but it is now the **impact** effect. The long-run effect is $\delta/(1-\rho)$ — so getting ρ wrong gets *that* wrong too.

What a firm effect does to a series



Two firms, same $\rho = 0.6$, same shocks in distribution — different u_i. Back-substitute the model into itself:

$$Y_{it} \;=\; \frac{u_i}{1-\rho} \;+\; \delta \sum_{j \geq 0} \rho^j D_{i,t-j} \;+\; \sum_{j \geq 0} \rho^j \varepsilon_{i,t-j}$$

- The firm effect shifts the **level** — not the volatility, not the persistence. The series look alike and sit at different heights.
- **Remember this picture.** An estimator that cannot tell *high because it was high last year* from *high because this firm is simply high* will get ρ wrong. Both usual estimators fail exactly here.

One rule settles every calculation

Read the back-substitution again: Y_{is} is assembled from u_i and from shocks **dated** s **and earlier** — nothing later. If ε_{it} is serially uncorrelated and unrelated to u_i, that gives one rule:

$$\boxed{\mathbb{E}[Y_{is}\,\varepsilon_{it}] = 0 \;\text{ whenever }\; t > s, \qquad \mathbb{E}[Y_{is}\,\varepsilon_{is}] = \sigma_\varepsilon^2}$$

- In words: **a later shock has nothing to do with an earlier** Y; the same-date shock enters Y with coefficient one.
- Every derivation in this lecture is a **date check** against this rule. Write out a product, look at the two dates, and either it is zero or it is σ_ε^2.
- One clause is doing quiet work: *serially uncorrelated*. If $\varepsilon_{i,t-1}$ predicted ε_{it}, a later shock *would* be related to an earlier Y. We will come back to that clause — it is the identifying assumption of the whole enterprise.

The problem you cannot assume away

Lag the model one period and the difficulty is on the page:

$$Y_{i,t-1} \;=\; \rho\, Y_{i,t-2} + \delta D_{i,t-1} + \underbrace{u_i}_{\text{also in today's error}} + \varepsilon_{i,t-1}$$

- Every endogenous regressor so far was endogenous because of something we **failed to measure**. This one is endogenous **by construction** — you can read it straight off the model.
- $Y_{i,t-1}$ **contains** u_i, and so does the error. Pooled OLS is biased.
- **Strict exogeneity is impossible.** ε_{it} moves Y_{it}, and Y_{it} *is* the regressor next period. The assumption fixed effects needs is contradicted by the model itself.
- So pooled OLS fails and fixed effects fails. The useful part, which takes the next section to show: **they fail in opposite directions.**

2. Two Estimators, Two Wrong Answers

Why OLS reads heterogeneity as persistence

Strip the model to the bone. Set $\rho = 0$ — **no persistence whatsoever**, only permanent differences between firms:

$$Y_{it} \;=\; u_i + \varepsilon_{it}$$

- A high-u_i firm has a high Y this year **and** had a high Y last year. Not because the past caused the present — because the firm is simply like that.
- Pooled OLS cannot tell those apart. It sees Y following Y and calls it persistence:

$$\operatorname{plim} \hat{\rho}_{\text{OLS}} \;=\; \frac{\operatorname{Cov}(Y_{it}, Y_{i,t-1})}{\operatorname{Var}(Y_{i,t-1})} \;=\; \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \;>\; 0 \;=\; \rho$$

- With $\sigma_u = \sigma_\varepsilon = 1$ that is 0.5. Simulated with $T = 5$ years and 100,000 firms, pooled OLS returns $\hat{\rho} = \mathbf{0.501}$ — when the truth is **zero**.
- **And it does not improve with T.** More years of the same firm being itself is more of the same evidence.

Why demeaning manufactures mean reversion

Fixed effects removes u_i by subtracting the firm mean — and the firm mean contains the errors.

$$\underbrace{Y_{i,t-1} - \bar{Y}_{i,-1}}_{\text{regressor}} \qquad \text{regressed on} \qquad \underbrace{\varepsilon_{it} - \bar{\varepsilon}_i}_{\text{error}}$$

- $\varepsilon_{i,t-1}$ pushes the **regressor up** — it is part of $Y_{i,t-1}$. It also pushes $\bar{\varepsilon}_i$ up, and $\bar{\varepsilon}_i$ is **subtracted** from the error. The two sides are forced apart mechanically:

$$\text{Cov}\left(Y_{i,t-1},\, -\bar{\varepsilon}_i\right) \;<\; 0, \qquad \text{of order } \sigma_\varepsilon^2/T$$

- So within a firm the data look **more mean-reverting than they are**, and $\hat{\rho}$ is biased **down**. This is **Nickell (1981) bias**.
- **The same $\rho = 0$ experiment, now estimated with fixed effects:** $\hat{\rho}_{\text{FE}} = -\mathbf{0.251}$, almost exactly $-1/(T-1)$. Pooled OLS invented persistence; demeaning invents **mean reversion**.
- The sting: this happens **even if u_i does not exist** — demeaning alone does it. And it fades only at rate $1/T$.

A derivation you can do at the board: $T = 3$

Take the smallest panel that admits the model: $T = 3$ years, **one differenced observation per firm**. Then $\hat{\rho}_{\mathsf{FE}}$ is exactly least squares on $\Delta Y_{i3} = \rho\, \Delta Y_{i2} + \Delta\varepsilon_{i3}$, and

$$\operatorname{plim}(\hat{\rho}_{\mathsf{FE}} - \rho) \;=\; \frac{\mathbb{E}[\Delta Y_{i2}\, \Delta\varepsilon_{i3}]}{\mathbb{E}[(\Delta Y_{i2})^2]}$$

- **Numerator, by the rule.** Expand $(Y_{i2} - Y_{i1})(\varepsilon_{i3} - \varepsilon_{i2})$ into four products and check dates. Three pair a Y with a *later* shock: zero. One pairs Y_{i2} with its *own* shock ε_{i2}, with a minus sign: $-\sigma_\varepsilon^2$.
- **Denominator, from the picture.** u_i cancels in a difference, so only the $\sum \rho^j \varepsilon$ part matters, with variance $\sigma_\varepsilon^2/(1-\rho^2)$ and one-year autocorrelation ρ. A difference of two such terms has variance $2(1-\rho)\cdot\sigma_\varepsilon^2/(1-\rho^2) = 2\sigma_\varepsilon^2/(1+\rho)$.

$$\operatorname{plim}(\hat{\rho}_{\mathsf{FE}} - \rho) \;=\; \frac{-\sigma_\varepsilon^2}{2\sigma_\varepsilon^2/(1+\rho)} \;=\; -\,\frac{1+\rho}{2}$$

- $-1/2$ at $\rho = 0$, approaching -1 as $\rho \to 1$. **For every $\rho < 1$ the plim of $\hat{\rho}_{\mathsf{FE}}$ is negative**: a firm whose leverage is genuinely 90% persistent is reported as mean-reverting. (Hansen 2022, §17.37.)

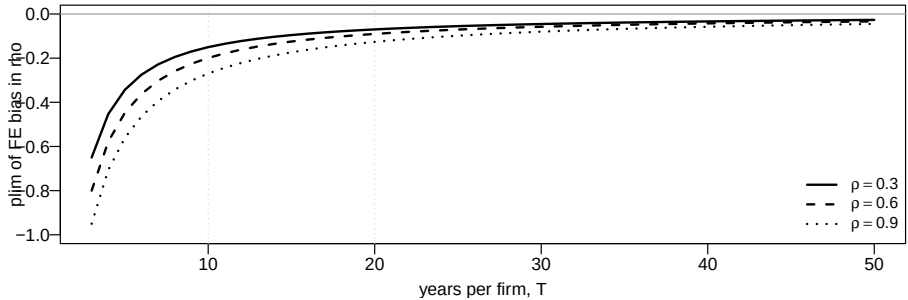
Watching both biases at once

Simulate $Y_{it} = \rho Y_{i,t-1} + u_i + \varepsilon_{it}$ with $\rho = 0.6$ and 4,000 firms, varying the number of years T.

Years per firm	$T = 3$	$T = 5$	$T = 10$	$T = 30$
Pooled OLS (biased **up**)	0.926	0.919	0.920	0.921
Fixed effects (biased **down**)	−0.181	0.148	0.399	0.543
Nickell (1981) plim, for comparison	*-0.200*	*0.153*	*0.401*	*0.542*
Truth	0.600	0.600	0.600	0.600

- **The OLS row never moves.** Its bias confuses heterogeneity with persistence, and more years cannot resolve it.
- **At $T = 3$ fixed effects reports a negative number** for a process whose true persistence is $+0.6$ — exactly what $-(1 + \rho)/2$ predicted.
- The simulated FE row tracks Nickell's plim throughout, so this is bias, not noise.

Does it fade with T? Slowly.



- The bias is $O(1/T)$: doubling the panel halves it. Finance panels are typically 10–20 years, and often unbalanced with much shorter firm histories.
- At $T = 30$ the bias is still -0.058 at $\rho = 0.6$ and -0.080 at $\rho = 0.9$. **The more persistent the series, the slower the bias leaves.**
- Hansen's verdict is blunt: *fixed effects should not be used for models with lagged dependent variables* **even when T is large.**

Two wrongs that bound the right answer

Neither estimator is consistent. But you know the **sign of each bias**, so together they fence the truth in:

$$\hat{\rho}_{\text{FE}} \;<\; \rho \;<\; \hat{\rho}_{\text{OLS}}$$

- In the table: $0.15 < 0.60 < 0.92$ at $T = 5$. The bracket is wide, and **honest**. A single number from either estimator would not be.
- **This is the standard first move in dynamic panel work** (Bond 2002). Run OLS and fixed effects *before* anything clever, and keep the two numbers. They are the goalposts.
- **Any estimator you trust must land between them.** We will build one in the next section; if it comes out above $\hat{\rho}_{\text{OLS}}$ or below $\hat{\rho}_{\text{FE}}$, you have learned something is wrong before interpreting a single coefficient.
- A bracket is not a confidence interval. It is a statement about **bias**, and says nothing about sampling noise.

Bracketing, the other kind

The same trick works when you are unsure **which model** is right — lagged Y, or fixed effects — and what you care about is δ.

If the truth is	and you estimate	then $\hat{\delta}$ is
the lagged model	fixed effects	too **high**
the fixed effects model	the lagged model	too **low**

As written, for $\delta > 0$. If $\delta < 0$ the two directions swap — the bracket still holds, but read it the other way round. (Angrist and Pischke 2009, §§5.3–5.4.)

- Run both and report the bracket rather than picking the specification you prefer. It is an honest statement of what the data support.
- **If the bracket excludes every sensible value — stop.** The model is wrong in some other way, or finite-sample bias is severe.

3. The Fix: Instruments from the Past

The fix, in one idea: Anderson–Hsiao

Difference the model to kill u_i:

$$\Delta Y_{it} \;=\; \rho\,\Delta Y_{i,t-1} + \delta\,\Delta D_{it} + \Delta\varepsilon_{it}$$

- It also **leaves a new problem behind**: $Y_{i,t-1}$ contains $\varepsilon_{i,t-1}$, and so does $\Delta\varepsilon_{it}$. Regressor and error share a shock, and you can say exactly how much:

$$\mathbb{E}\big[\Delta Y_{i,t-1}\,\Delta\varepsilon_{it}\big] \;=\; -\,\sigma_\varepsilon^2$$

- **So go back one more lag.** $Y_{i,t-2}$ is **far enough back to be clean** — it was already determined before either $\varepsilon_{i,t-1}$ or ε_{it} happened — yet **close enough to still predict** $\Delta Y_{i,t-1}$:

$$\mathbb{E}\big[Y_{i,t-2}\,\Delta\varepsilon_{it}\big] \;=\; 0$$

- That is **Anderson–Hsiao (1982)**: IV on the differenced equation using $Y_{i,t-2}$. It needs $T \geq 3$.
- Note what has happened. The lagged outcome caused the problem, and a *longer* lag of the same variable solves it — no new data, only the history you already have.
- Those two lines carry the section. **The next two slides are the arithmetic behind them.**

Why differencing alone fails: four terms, one survivor

Expand $\mathbb{E}\big[(Y_{i,t-1} - Y_{i,t-2})(\varepsilon_{it} - \varepsilon_{i,t-1})\big]$ and check the dates against the rule:

Term	Shock dated	Value
$+\,\mathbb{E}[Y_{i,t-1}\varepsilon_{it}]$	after the Y	0
$-\,\mathbb{E}[Y_{i,t-1}\varepsilon_{i,t-1}]$	**same as the Y**	$-\sigma_\varepsilon^2$
$-\,\mathbb{E}[Y_{i,t-2}\varepsilon_{it}]$	after	0
$+\,\mathbb{E}[Y_{i,t-2}\varepsilon_{i,t-1}]$	after	0

- One term survives — the firm's **own shock**, which pushes the regressor up and is *subtracted* out of the error. Exactly $-\sigma_\varepsilon^2$, shrinking in neither N, T, nor ρ.
- This is the same calculation as the $T = 3$ numerator two sections ago. Differencing is not a fix; it is where the fixed-effects bias **comes from**.
- Compare the regressor to a hypothetical: if we could use $Y_{i,t-2}$ in place of $\Delta Y_{i,t-1}$, the surviving row would read "after", and the whole table would be zero. That is the next slide.

Why $Y_{i,t-2}$ is clean — and still useful

The same rule, applied twice:

$$\mathbb{E}\big[Y_{i,t-2}\,\Delta\varepsilon_{it}\big] = \underbrace{\mathbb{E}[Y_{i,t-2}\,\varepsilon_{it}]}_{t>t-2\,\Rightarrow\,0} - \underbrace{\mathbb{E}[Y_{i,t-2}\,\varepsilon_{i,t-1}]}_{t-1>t-2\,\Rightarrow\,0} = 0$$

- Both shocks are now in the instrument's future, so both terms die. The only term that ever survived was the same-date one, and stepping back a single period is precisely what removes it. **The difference between a broken regressor and a valid instrument is one date.**
- **Relevance, from the same picture.** u_i cancels inside $\Delta Y_{i,t-1}$, so only the $\sum \rho^j \varepsilon$ part of $Y_{i,t-2}$ can covary with it:

$$\mathrm{Cov}\,(Y_{i,t-2},\,\Delta Y_{i,t-1}) = (\rho-1)\,\frac{\sigma_\varepsilon^2}{1-\rho^2} = -\frac{\sigma_\varepsilon^2}{1+\rho} \neq 0$$

- So the instrument is relevant and the IV is identified. Keep both numbers. The first-stage coefficient is that covariance over $\mathrm{Var}(Y_{i,t-2}) = \sigma_u^2/(1-\rho)^2 + \sigma_\varepsilon^2/(1-\rho^2)$ — and **that ratio is where the fix falls apart** (Section 4).

Two assumptions doing all the work

- **Validity.** $\mathbb{E}[Y_{i,t-2}\Delta\varepsilon_{it}] = 0$ holds *only* because ε_{it} is serially uncorrelated — the quiet clause in the rule. Which is to say, **only because the dynamics are correctly specified**. If the truth is AR(2) and you fit AR(1), the leftover dynamics land in ε_{it}; now $\varepsilon_{i,t-2}$ (inside $Y_{i,t-2}$) predicts $\varepsilon_{i,t-1}$, and the instrument is contaminated.
- So the lag length is not a specification detail you tune at the end. It is the **identifying assumption**. It is testable, and we will test it.
- **Relevance.** On paper it holds — we showed $\mathrm{Cov}(Y_{i,t-2}, \Delta Y_{i,t-1}) = -\sigma_\varepsilon^2/(1+\rho) \neq 0$. What matters in practice is the first-stage *coefficient*, and that collapses in exactly the cases finance cares about.
- The same two assumptions, applied to D: if D is **strictly exogenous**, ΔD_{it} is its own instrument. If D is only **predetermined** — responds to past shocks but not to today's — then $D_{i,t-1}, D_{i,t-2}, \ldots$ are valid instruments for ΔD_{it}, by the same date check.

Arellano–Bond: use every lag you have

Anderson–Hsiao uses one moment condition out of many. If ε_{it} is serially uncorrelated, then at date t **every** Y dated $t-2$ or earlier passes the date check (Holtz-Eakin, Newey and Rosen 1988; Arellano and Bond 1991):

Differenced equation at	valid instruments for $\Delta Y_{i,t-1}$	how many
$t=3$	Y_{i1}	1
$t=4$	$Y_{i1},\ Y_{i2}$	2
$t=5$	$Y_{i1},\ Y_{i2},\ Y_{i3}$	3
$\vdots$		
$t=T$	$Y_{i1},\ \ldots,\ Y_{i,T-2}$	$T-2$

- Each period gets its own block: a **block-diagonal** instrument matrix with $(T-1)(T-2)/2$ columns. Overidentified once $T>3$, so the estimator is **GMM** rather than IV.
- **Gain:** efficiency and stability — one-instrument Anderson–Hsiao can be very noisy. **Cost:** the count grows with T^2: 91 instruments at $T=15$, 406 at $T=30$. Next slide.

Too many instruments

Roodman (2009): a GMM estimator with many instruments relative to N is quietly a bad estimator.

- **It overfits the endogenous regressor.** With enough instruments, the first stage can reproduce $\Delta Y_{i,t-1}$ almost exactly — including the part correlated with $\Delta\varepsilon_{it}$. The estimator drifts back toward the biased OLS answer it was built to avoid.
- **It disarms the overidentification test.** The Hansen J statistic becomes too weak to reject anything; a p-value of 0.99 is a warning sign, not a clean bill of health.
- **Two remedies, both standard.** *Limit the lag depth* — use $Y_{i,t-2}$ and $Y_{i,t-3}$ only, say. *Collapse* — replace the block-diagonal matrix by one column per lag, so the count grows with T instead of T^2.
- Rule of thumb: keep the instrument count **well below** N, and **report it**. A referee who cannot see the count cannot evaluate the estimate.

One step, two steps, and the standard errors

GMM needs a weighting matrix, and the choice matters in small samples.

- **One-step** uses a weighting matrix built from the known structure of $\Delta\varepsilon$ — adjacent differences share a shock, so they are correlated $-\frac{1}{2}$. It does not depend on residuals, so it is *less random*. Fine under homoskedasticity.
- **Two-step** re-weights using first-step residuals. Asymptotically efficient under heteroskedasticity — but its standard errors are **badly biased downward** in finite samples, because they ignore the noise in the estimated weights.
- **Windmeijer (2005)** corrects the two-step standard errors, and the correction is large. Stata applies it by default with `twostep vce(robust)`. Never report uncorrected two-step standard errors.
- Practical reading: with a few hundred firms and a dozen years, one-step with robust standard errors and two-step with Windmeijer should agree. If they do not, something is weak.

Diagnostics you must report

Every one of these is a check on an assumption the estimator cannot verify for itself.

- **AR(1) in the differenced residuals — expected.** $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-1}$ share $\varepsilon_{i,t-1}$, so first-order correlation of about $-\frac{1}{2}$ is a sign the model is behaving.
- **AR(2) in the differenced residuals — must be absent.** $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-2}$ share nothing *unless* ε *itself is serially correlated*. If it is, $Y_{i,t-2}$ fails the date check and **every instrument is invalid**. This is the test of the identifying assumption. It is not optional.
- **Hansen J test of overidentifying restrictions** — with the instrument count beside it, for the reason on the previous slide.
- **The bracket.** Does the estimate sit between $\hat{\rho}_{\text{FE}}$ and $\hat{\rho}_{\text{OLS}}$? If it drifts toward the OLS end, suspect too many instruments; if toward the FE end, suspect weak ones.
- **The first stage**, as for any IV. The next section explains why it is usually the problem.

4. Why the Fix Can Fail

Weak instruments: the first stage, in closed form

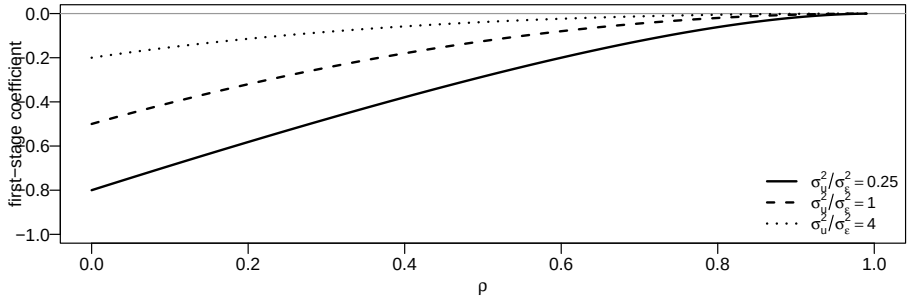
Blundell and Bond (1998) wrote the first stage, $\Delta Y_{i,t-1} = \gamma\, Y_{i,t-2} + v_{it}$, and divided the two quantities from Section 3:

$$\gamma \;=\; \frac{\mathrm{Cov}(Y_{i,t-2}, \Delta Y_{i,t-1})}{\mathrm{Var}(Y_{i,t-2})} \;=\; \frac{-\sigma_\varepsilon^2/(1+\rho)}{\sigma_u^2/(1-\rho)^2 + \sigma_\varepsilon^2/(1-\rho^2)}$$

$$\;=\; (\rho - 1)\,\frac{k}{k + \sigma_u^2/\sigma_\varepsilon^2}, \qquad k = \frac{1-\rho}{1+\rho}$$

- The covariance on top never goes to zero; **the damage is all in the denominator.** The *level* $Y_{i,t-2}$ gets swamped by things that say nothing about the *change* $\Delta Y_{i,t-1}$.
- **As** $\rho \to 1$, the level wanders without bound while changes stay the same size — a random walk's level says nothing about its next step.
- **As** $\sigma_u^2/\sigma_\varepsilon^2$ **grows**, the level is mostly the firm's permanent height, which cancels out of the change. In the two-firm picture: the heights tell you nothing about the wiggles.
- Both are the normal case in corporate finance: **persistent series, large firm effects.**

A picture of the first stage



- At $\rho = 0.6$ with equal variances, $\gamma = -0.08$: a usable first stage. At $\rho = 0.95$, $\gamma = -0.0013$ — essentially nothing.
- Leverage ratios, payout ratios and cash holdings all sit in the right-hand third of this plot. **The instrument is weakest exactly where you most need it.**

Watching it fail

Same simulation design, now with four estimators. 500 firms, $T = 6$, 200 replications; the table reports the **median** and the **10th–90th percentile range** across replications.

	$\rho = 0.6$	$\rho = 0.95$
Pooled OLS	0.92 [0.91, 0.93]	1.00 [1.00, 1.00]
Fixed effects	0.23 [0.20, 0.26]	0.47 [0.44, 0.50]
Anderson–Hsiao (one instrument)	0.59 [0.50, 0.71]	0.86 [−0.41, 2.16]
Arellano–Bond (all lags, one-step)	0.58 [0.51, 0.67]	0.55 [0.19, 0.93]

- **At $\rho = 0.6$ the fix works.** Both IV estimators sit inside the bracket, close to the truth; Arellano–Bond is the tighter of the two.
- **At $\rho = 0.95$ it does not.** Anderson–Hsiao is scattered across the number line. Arellano–Bond is biased **down toward fixed effects** with an enormous spread — the textbook signature of weak instruments.

System GMM: the idea, and its price

Blundell and Bond's repair: keep the differenced equation, and **add the levels equation back**, instrumented the other way round:

$$Y_{it} = \rho\, Y_{i,t-1} + u_i + \varepsilon_{it}, \qquad \text{instrument } Y_{i,t-1} \text{ with } \Delta Y_{i,t-1}$$

- **Why it helps.** The change predicts the level even when the level cannot predict the change: $\mathrm{Cov}(\Delta Y_{i,t-1}, Y_{i,t-1})$ stays bounded away from zero as $\rho \to 1$. The first stage no longer dies.
- **What it costs.** The levels error still contains u_i, so you need $\mathbb{E}[\Delta Y_{i,t-1}\, u_i] = 0$: **changes must be unrelated to the firm's permanent level.** That is exactly what the two-firm picture shows — *if both firms started at their own long-run levels*.
- The formal name is **mean stationarity**: a firm's distance from its own long-run level in year 1 is unrelated to that level.
- **When to doubt it.** Firms observed from IPO are typically *below* long-run leverage and converging up; a panel that starts after a crisis has everyone off target in a correlated way. There the extra instruments are invalid, and system GMM rests on an assumption you do not believe. Say which case you are in.

Long panels: a different route

When T is large the problem is milder and a different tool is available.

- The fixed-effects bias is $O(1/T)$ **and its formula is known** (Nickell 1981). So correct it: estimate FE, compute the bias at $\hat{\rho}$, subtract, iterate. Kiviet (1995); Bruno (2005) for unbalanced panels — Stata's `xtlsdvc`. No instruments, no weak-instrument problem.
- Hahn and Kuersteiner (2002) show the corrected estimator is efficient as $N, T \to \infty$ together; Hahn, Hausman and Kuersteiner (2007) use **long differences** — $Y_{it} - Y_{i,t-k}$ — which keeps the instruments strong even for persistent series.
- Flannery and Hankins (2013) ran a horse race on finance-like panels — short, persistent, unbalanced, with endogenous controls — and found the bias-corrected FE and system GMM estimators generally reliable, and difference GMM frequently not.
- **Rule of thumb.** $T \leq 10$ and ρ moderate: Arellano–Bond, with the diagnostics. ρ near one: system GMM, stating the initial-condition assumption. $T \geq 20$: bias-corrected FE, which has fewer ways to go wrong.

5. In Practice

A case study: how fast do firms adjust leverage?

The partial-adjustment model is our equation with $Y =$ leverage and $1 - \rho$ the **speed of adjustment**. The literature's estimates:

Paper	Estimator	$\hat{\rho}$	speed $1 - \hat{\rho}$
Fama and French (2002)	pooled (Fama–MacBeth)	≈ 0.85	$\approx$ 7–18% a year
Flannery and Rangan (2006)	fixed effects	≈ 0.66	$\approx$ 34%
Lemmon, Roberts and Zender (2008)	system GMM	≈ 0.75	$\approx$ 25%
Huang and Ritter (2009)	long differencing	≈ 0.80	$\approx$ 17–23%

- A decade of disagreement about whether firms have targets at all — and **the ordering is the bracket**. Fixed effects is biased down in ρ, so it reports the *fastest* adjustment; pooled estimation is biased up, so the *slowest*. The IV-type estimates sit between.
- The substantive question (do firms rebalance toward targets, or does leverage just drift?) **turned on the bias this lecture is about.** A half-life of 1.7 years and one of 5 years are different theories of the firm.

Try it: simulate the panel

Truth known, so every estimator can be graded. Full do-file: `06b_dynamic_panel_sim.do`.

```
clear all
set seed 20260202
local N = 500            // firms
local T = 8              // years kept per firm
local rho = 0.6          // true persistence -- change to 0.95 later
set obs `N'
gen firm = _n
gen u = rnormal()                                        // firm effect
gen y = u/(1-`rho') + rnormal(0, 1/sqrt(1-`rho'^2))      // stationary start
forvalues s = 1/30 {
    replace y = `rho'*y + u + rnormal()                  // burn in to the long-run level
}
expand `T'
bysort firm: gen year = _n
xtset firm year
bysort firm (year): replace y = `rho'*y[_n-1] + u + rnormal() if _n > 1
```

- The burn-in starts every firm at its own long-run level — the **mean-stationarity** condition system GMM needs. Delete it, and system GMM is misspecified.

Try it: estimate, grade, and watch it fail

```
reg   y L.y, vce(cluster firm)                          // biased up
estimates store OLS
xtreg y L.y, fe vce(cluster firm)                       // biased down (Nickell)
estimates store FE
ivregress 2sls D.y (LD.y = L2.y), vce(cluster firm)  // Anderson-Hsiao
estimates store AH
xtabond y, lags(1) maxldep(3) twostep vce(robust)    // Arellano-Bond, lags 2-4
estimates store AB
estat abond                                             // AR(1) expected, AR(2) must be absent
xtdpdsys y, lags(1) maxldep(3) twostep vce(robust)   // Blundell-Bond system GMM
estimates store BB
estimates table OLS FE AH AB BB, keep(L.y LD.y) b(%7.3f) se(%7.3f)
```

- **What to expect at $\rho = 0.6$:** OLS near 0.92, fixed effects near 0.34 (Nickell's plim at $T = 8$), and the three IV/GMM estimates near 0.6 — inside the bracket, with Anderson–Hsiao the noisiest.
- **Then set $\rho = 0.95$ and run it again.** Anderson–Hsiao scatters; Arellano–Bond drifts down toward fixed effects with a wide standard error; system GMM holds up — because the burn-in made its assumption true. That is the whole lecture in one table.

The code, on real data

```
* 1. The bracket first -- and keep both numbers
reg   y L.y x i.year, cluster(firm)
xtreg y L.y x i.year, fe cluster(firm)

* 2. Difference GMM (Arellano-Bond): two-step, Windmeijer SEs, lag depth capped at 3
xtabond y x i.year, lags(1) maxldep(3) twostep vce(robust)
estat abond                          // AR(1) expected; AR(2) must be absent

* 3. Same, with collapsed instruments and the Hansen test (Roodman's xtabond2)
ssc install xtabond2
xtabond2 y L.y x i.year, gmm(L.y, lag(2 4) collapse) iv(x i.year) twostep robust

* 4. System GMM -- only after you have stated the initial-condition assumption
xtdpdsys y x i.year, lags(1) maxldep(3) twostep vce(robust)

* 5. Long T: bias-corrected fixed effects
ssc install xtlsdvc
xtlsdvc y x, initial(ab) bias(3) vcov(50)
```

- In R: `plm::pgmm()` with `transformation = "d"` (difference) or `"ld"` (system); `pdynmc` for the Roodman-style options.

A reporting checklist

Report	Because
$\hat{\rho}_{\text{OLS}}$ and $\hat{\rho}_{\text{FE}}$	the bracket; your estimate must sit inside it
AR(1) and AR(2) tests	AR(2) is the identifying assumption
Hansen J *and* the instrument count	the test is only informative when the count is small relative to N
lag depth, and whether collapsed	so the reader can judge the count
one-step and two-step (Windmeijer) side by side	disagreement signals weakness
difference vs. system GMM, with the initial-condition assumption stated	they are identified by different assumptions
what D is assumed to be (exogenous / predetermined / endogenous)	it determines which lags of D are valid instruments

- None of this is decoration. Each row is a place where the estimator can silently fail, and each is a number the reader needs to decide whether it did.

Conclusion

1. **A lagged Y is endogenous by construction**, not by omission. You can read the problem off the model.
2. **One rule does every calculation**: later shocks are unrelated to earlier Y; the same-date shock enters with coefficient one. Check dates.
3. **OLS is biased up and fixed effects down.** Neither improves with N; FE improves only at rate $1/T$. Run both first — **the bracket is the goalposts.**
4. **The fix is a longer lag of the same variable.** $Y_{i,t-2}$ is one date too early to be contaminated. Arellano–Bond uses all such lags; cap and collapse them.
5. **The fix fails when the series is persistent or firm effects are large** — the normal case in finance. Then: system GMM with its assumption stated, or bias-corrected FE if T allows.

- The question to carry out: **which way is this estimator wrong?** If you can say that, you can say what the truth is between.
- **Next:** difference-in-differences — when the switching has a *date*.

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